

CLINICAL INVESTIGATION

Eye

PROTON BEAM RADIOTHERAPY VERSUS FRACTIONATED STEREOTACTIC RADIOTHERAPY FOR UVEAL MELANOMAS: A COMPARATIVE STUDY

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Purpose: A comparative treatment planning study was undertaken between proton and photon therapy in uveal melanoma to assess the potential benefits and limitations of these treatment modalities. A fixed proton horizontal beam (OPTIS) and intensity-modulated spot-scanning proton therapy (IMPT), with multiple noncoplanar beam arrangements, was compared with linear accelerator-based stereotactic radiotherapy (SRT), using a static and a dynamic micromultileaf collimator and intensity-modulated RT (IMRS).

Method and Materials: A planning CT scan was performed on a brain metastasis patient, with a 3-mm acquisition slice spacing and the patient looking at a luminous spot with the eyes in three different positions (neutral and 25° right and left). Four different gross tumor volumes were defined for each treatment technique. These target scenarios represented different locations (involving vs. not involving the macula and temporal vs. nasal) and volumes (10 × 6 mm vs. 16 × 10 mm) to challenge the proton and photon treatment techniques. The planning target volume was defined as the gross tumor volume plus 2 mm laterally and 3 mm craniocaudally for both modalities. A dose homogeneity of 95–99% of the planning target volume was used as the “goal” for all techniques. The dose constraint (maximum) for the organs at risk (OARs) for both the proton and the SRT photon plans was 27.5, 22.5, 20, and 9 CGE-Gy for the optic apparatus, retina, lacrimal gland, and lens, respectively. The dose to the planning target volume was 50 CGE-Gy in 10 CGE-Gy daily fractions. The plans for proton and photon therapy were computed using the Paul Scherrer Institute and BrainSCAN, version 5.2 (BrainLAB, Heimstetten, Germany) treatment planning systems, respectively. Tumor and OARs dose–volume histograms were calculated. The results were analyzed using the dose–volume histogram parameters, conformity index (CI_{95%}), and inhomogeneity coefficient.

Results: Target coverage of all simulated uveal melanomas was equally conformal with the photon and proton modalities. The median CI_{95%} value was 1.74, 1.86, and 1.83 for the static, dynamic, and IMSRT plans, respectively. With proton planning, the median CI_{95%} was 1.88 for OPTIS and substantially improved with IMPT in some tumor cases (median CI_{95%}, 1.29). The tumor dose homogeneity in the proton plans was, however, always better than with SRT photon planning (median inhomogeneity coefficient 0.1 and 0.15 vs. 0.46, 0.41, and 0.23 for the OPTIS and IMPT vs. the static, dynamic, and IMSRT plans, respectively). Compared with the photon plans, the use of protons did not lead to a substantial reduction in the homolateral OAR total integral dose in the low- to high-dose level, except for the lacrimal gland. The median maximal dose and dose at the 10% volume with the static, dynamic, and IMSRT plans was 33–30.8, 31.8–28, and 35.8–49 Gy, respectively, for the lacrimal gland, a critical organ. For protons, only the OPTIS plans were better, with a median maximal dose and dose at the 10% volume using OPTIS and IMPT of 19.2 and 8.8 and 25.6 and 23.6 CGE, respectively. The contralateral OARs were completely spared with the proton plans, but the median dose delivered to these structures was 1.2 Gy (range, 0–6.3 Gy) with the SRT photon plans.

Conclusion: These results suggest that the use of SRT photon techniques, compared with protons, can result in similar levels of dose conformation. IMPT did not increase the degree of conformality for this small tumor. Tumor dose inhomogeneity was, however, always increased with photon planning. Except for the lacrimal gland, the use of protons, with or without intensity modulation, did not increase homolateral OAR dose sparing. The dose to all the contralateral OARs was, however, completely eliminated with proton planning. © 2005 Elsevier Inc.

Proton beam therapy, Stereotactic radiotherapy, Intensity-modulated radiotherapy, Uveal melanoma, Dose-comparative study, Conformity index, Inhomogeneity coefficient.